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Multi-hop graph structural modeling for cancer-related circRNA-miRNA interaction prediction

Mengmeng Wei [a,1], Lei Wang [a,1,*], Xiaorui Su [b], Bowei Zhao [b], Zhuhong You [c,*]

[a] School of Computer Science and Technology, China University of Mining and Technology, Xuzhou, 221116, China
[b] Xinjiang Technical Institute of Physics and Chemistry, Chinese Academy of Sciences, Urumqi, 830011, China
[c] School of Computer Science, Northwestern Polytechnical University, Xi'an, 710129, China

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ABSTRACT

A substantial body of research indicates that circRNA can act as a sponge to absorb miRNA, thereby regulating the development of cancers. Existing circRNA-miRNA interactions (CMIs) prediction models mainly focus on single features and local structures of molecules, making it difficult to fully describe the overall properties of molecules and overlooking the multi-hierarchical associations between them. To address these challenges, we propose a computational model named GraCMI based on multi-hop graph structural modeling, which predicts CMIs by integrating structural and attribute information of molecules. GraCMI learns the representation of molecules in multi-level neighborhoods through constructing heterogeneous networks and performing high- and low-order matrix factorization. GraCMI captures both the intrinsic properties and global structures of molecules, extracting and fusing multi-source features, improving prediction accuracy. In the case studies, 7 out of the top 10 CMI pairs predicted using GraCMI on a real cancer-related dataset were confirmed. Additionally, GraCMI demonstrates a competitive advantage on two other classic datasets. Overall, the experimental results show that GraCMI can effectively predict CMIs, which is expected to provide new insights into future miRNA-mediated circRNA regulation of cancer development.

1. Introduction

Non-coding RNAs (ncRNAs) play a crucial role in identifying and regulating potential targets in cancer [1]. MicroRNA (miRNA) is a type of ncRNA molecule, approximately 21–23 nucleotides in length [2]. Extensive research has confirmed that miRNAs from exosomes can serve as biomarkers for tumor monitoring, offering promising applications in diagnosis and treatment [3]. Circular RNA (circRNA) was first discovered in 1976 and is a type of ncRNA with a complete closed-loop structure [4]. Limited by traditional RNA detection methods, researchers believed these circRNAs might have been of viral origin and did not trigger in-depth research. However, since 2010, research on circRNAs has grown rapidly, and a large amount of evidence shows that they widely exist in the transcriptome of eukaryotes and play essential functions in various cellular processes [5]. With the continuous advancement of RNA-seq and computer technology, researchers have discovered that circRNAs can serve as miRNAs sponge to participate in post-transcriptional gene regulation [6]. For example, the expression levels of CircCD44 are significantly upregulated in triple-negative breast cancer (TNBC), and its expression is negatively correlated with the prognosis of TNBC patients [7]. CircCD44 acts as a sponge for miR-502-5p, interacting with IGF2BP2, a binding protein, thereby promoting proliferation, migration, invasion, and tumorigenesis of TNBC to a certain extent. In hepatocellular carcinoma (HCC), the knockdown of circUBE2J2 can lead to the downregulation of the miR-370-5p target, KLF7, thereby promoting the proliferation and migration of HCC cells [8]. The circRNA-miRNA interactions (CMIs) are key not only in cancer research but also in other disease areas, including myocardial infarction, coronary artery heart disease, and diabetic cardiomyopathy [9]. Additionally, some synthetic circRNAs are engineered to treat diseases by regulating miRNAs [10]. Therefore, accurately predicting potential CMIs has significant clinical implications.

In recent years, methods proposed for predicting CMIs have primarily been divided into experimental and computational approaches. While experimental methods can capture more detailed information and yield more accurate results, they are often costly and time-consuming.

* Corresponding authors.
E-mail addresses: a591286260@gmail.com (M. Wei), leiwang@cumt.edu.cn (L. Wang), suxiaorui19@mails.ucas.ac.cn (X. Su), zhaobowei19@mails.ucas.edu.cn (B. Zhao), zhuhongyou@nwpu.edu.cn (Z. You).

[1] The first two authors contribute equally.

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With the rapid advancement of computer performance and machine learning algorithms, computational methods can swiftly predict high-probability potential interaction pairs, providing reliable candidates for experiments [11]. Computational methods are typically categorized into three types, matrix-based methods, deep learning-based methods, and graph embedding-based methods. Matrix-based methods use known CMIs to calculate similarity and construct characterization matrices [12]. Potential molecular features are extracted through matrix operations, and the final output matrix is the prediction result. These methods have low computational resource requirements but struggle to capture intricate nonlinear relationships. Deep learning-based methods perform well when dealing with large-scale datasets, possessing strong fitting capabilities [13]. However, deep learning-based methods demand higher computational performance and have relatively complex working principles with comparatively poorer interpretability. Graph embedding-based methods have been proven effective in handling topological information, and they map nodes in the graph to a low-dimensional vector space, obtain feature representations of molecules, and apply them to prediction tasks [14]. These methods typically focus on capturing structural information within the graph and may overlook the intrinsic features of nodes.

Although the aforementioned computational methods have made certain progress, there are still the following challenges in CMI prediction. (1) The nodes in graph-structured data exhibit intricate interaction patterns, wherein target nodes are typically interconnected with multi-order neighbors, making it challenging to accurately quantify and represent the influence propagated from higher-order neighboring nodes. (2) While graph embedding-based approaches have demonstrated their prowess in extracting structural features, the intrinsic features of nodes themselves are often relegated to the background. (3) During node feature extraction, incomplete data or computational errors may introduce noise interference, potentially degrading the quality of the learned representations. Therefore, predicting unknown CMIs from the perspective of molecular structure and inherent features, combined with noise reduction algorithms, can improve prediction accuracy.

In this paper, we propose GraCMI, a noise-reduction model for CMI prediction based on multi-hop graph structures and multi-source feature fusion. As illustrated in Fig. 1, we first construct a heterogeneous network by combining the known associations among circRNAs, miRNAs, and cancers. Next, molecular functional similarity is obtained by calculating the Gaussian interaction profile kernel (GIPK) and Jaccard similarity between nodes of the same type. These similarity matrices are integrated and subjected to denoising reconstruction through a Sparse Autoencoder (SAE) to derive the molecular attribute features [15]. The graph embedding method Grarep is employed to learn the multi-hop graph structural features of nodes within the heterogeneous network [16]. Finally, the attribute features and structural features of molecules are fused and input into the eXtreme Gradient Boosting (XGBoost) classifier for CMIs prediction. Experimental results demonstrate that GraCMI can effectively predict potential relationships between circRNAs and miRNAs associated with cancers.

In summary, the main contributions of this paper are as follows:

- We collected a real-world dataset of CMIs associated with cancer, which has been experimentally validated to ensure its reliability and effectiveness.
- GraCMI reconstructs the molecular attribute features, reducing noise and improving the quality of feature representations.
- GraCMI leverages a multi-hop mechanism to characterize the global information of the graph, extracting structural features and enhancing the model's ability to capture complex relationships.
- By fusing molecular multi-source features, GraCMI enables accurate prediction of unknown potential relationships, offering novel insights and potential applications in gene regulation and personalized medicine.

2. Related work

2.1. CMI prediction

In recent years, a series of computational models have been proposed for recognizing CMI. Early methods mainly focused on constructing shallow features and similarity calculations, which struggled to capture the complex structural information within networks. With the development of graph neural networks (GNNs), more and more studies have begun to introduce graph learning mechanisms to explore dependencies between nodes. For example, CMASG combines GNNs with singular value decomposition to learn node representations from both nonlinear and linear perspectives, achieving efficient prediction [17]. Additionally, some studies have introduced a weight mechanism in the propagation paths to enhance the recognition performance of candidate molecular pairs. KRWRMC is based on the random walk mechanism and incorporates

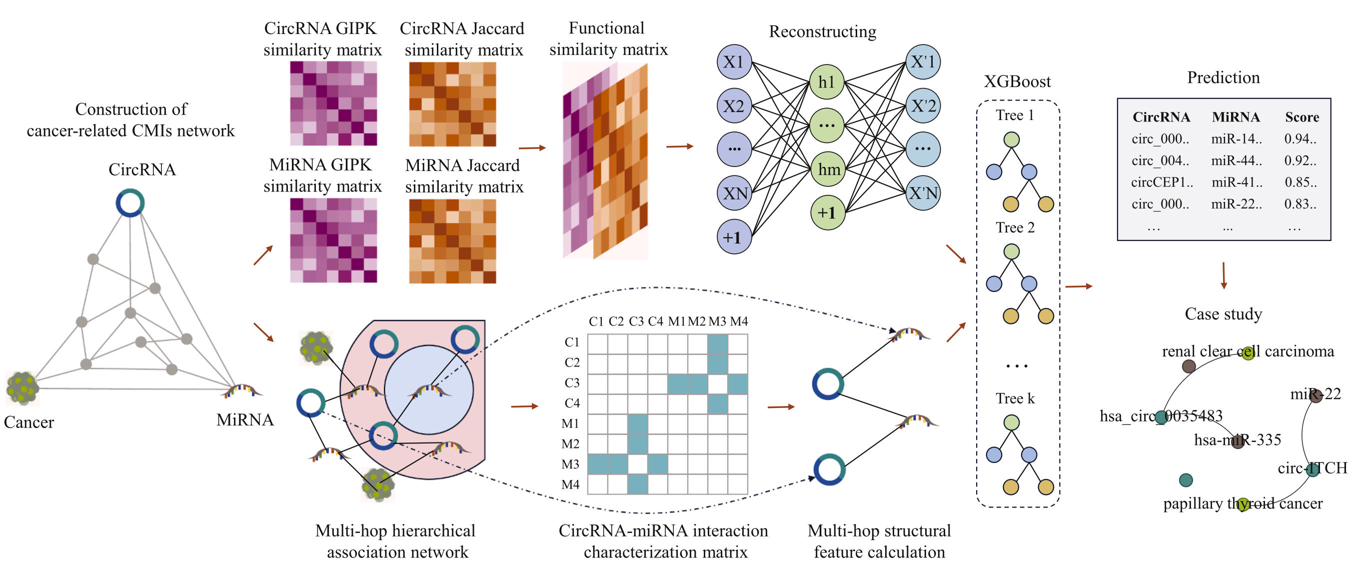

Fig. 1. Overview of GraCMI. First, molecular similarity is computed to obtain intrinsic attribute features, and a sparse autoencoder is applied to perform denoising through feature reconstruction. Then, multi-hop modeling is performed on the CMI network to extract global structural features. Finally, the denoised attribute features and structural features are integrated to predict potential CMIs.

a neighbor weighting design that emphasizes the importance of high-quality neighboring nodes [12]. By integrating a multi-label K-nearest neighbor strategy, it further optimizes the structural representation of nodes, thereby enhancing its ability to capture complex relationships within the network. NGCICM combines graph attention networks with induced matrix decomposition and introduces conditional random fields to extract contextual information, improving the ability to express complex interaction relationships [18].

Overall, current research has advanced the inferential capacity of models to uncover latent associations within CMI networks by integrating diverse network embedding methods and feature augmentation strategies. However, the majority of existing approaches remain dependent on local neighborhood information, lacking comprehensive modeling of cross-node path dependencies [19]. Therefore, constructing a more generalized and robust representation learning framework that fully leverages the latent dependencies among distant nodes remains a critical avenue for future investigation.

2.2. Multi-source feature fusion

With the increasing demand for modeling complex molecular interactions, multi-source information fusion has emerged as a pivotal strategy for enhancing the performance of CMI prediction. JSNDCMI fuses functional similarity and local structural information, employing denoising autoencoders to reconstruct multi-source features, thereby mitigating the impact of noise in sparse graphs [14]. BCMCMI employs a Bidirectional Encoder Representation from Transformers (BERT) to extract semantic features from molecular sequences [20]. By integrating structural features captured by Metapath2vec on a similarity-based heterogeneous graph, it effectively learns the topological information embedded in various path patterns along meta-paths. While the above methods have made progress in feature representation, the original biological attributes of molecules are prone to noise interference during encoding. The lack of a targeted denoising mechanism may lead to features deviating from their true distribution, affecting the prediction accuracy of the model.

This study proposes a novel framework GraCMI that combines multi-hop structural modeling with targeted denoising of biological attribute features, two aspects that are seldom explored by existing CMI prediction models. Unlike prior works relying on neighbor propagation, GraCMI explicitly models long-range dependencies through multi-order matrix factorization, embedding global structural information to capture richer topological features. To further enhance representation learning, GraCMI incorporates a denoising mechanism to preserve the fidelity of molecular attributes. By fusing topological structure with denoised attribute features, GraCMI achieves a more comprehensive and reliable molecular representation for downstream prediction tasks.

3. Materials and methods

3.1. Materials

In this study, we constructed a dataset from the CircR2cancer database, which contains 1135 circRNAs and 1439 associations linked to 82 types of cancer [21]. These data have undergone experimental validation and can be cross-referenced in relevant literature. After screening duplicate and invalid data, we obtained a final dataset containing 755 interaction pairs between 514 circRNAs and 471 miRNAs associated with 53 cancers. In addition, CMI-9589 and CMI-9905, as golden datasets in the field of CMIs prediction, were also used to verify the proposed model. The CMI-9589 dataset comes from the CircBank database, which was constructed by Liu et al. [22]. They utilize two distinct algorithms, Miranda and TargetScan, to predict the targeting relationships between circRNAs and miRNAs. We selected 9589 CMI pairs with higher confidence, encompassing 2115 circRNAs and 821 miRNAs.

Table 1
Statistics of three datasets.

Dataset	CircRNA	MiRNA	Interactions	Cancer	AVG-D
CMI-755	514	471	755	53	1.53
CMI-9589	2115	821	9589	N/A	6.53
CMI-9905	2346	962	9905	N/A	5.99

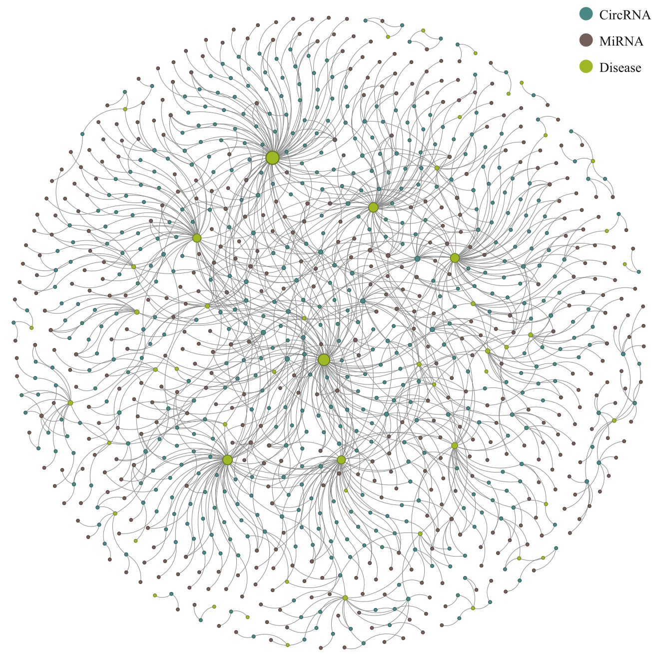

Fig. 2. Visualization of the CMI-755 dataset.

The CMI-9905 dataset combines data from the CircBank and CircR2Cancer databases, which contain 9905 pairs of CMI relationships involving 2346 circRNAs and 962 miRNAs [14]. The details of the three datasets are shown in Table 1, where AVG-D represents the average degree of the data, which reflects the average density of connections between nodes in the graph.

As shown in Table 1, CMI-9589 and CMI-9905 exhibit higher average degrees, which means there are more neighbors around the node. In contrast, CMI-755 dataset contains fewer nodes with fewer neighbors, and its visual representation is shown in Fig. 2. To reasonably construct the negative sample set, GraCMI randomly pairs circRNAs and miRNAs that are not found to be associated in the positive samples. We selected negative samples equal to the number of positive samples to avoid imbalance problems.

3.2. Molecular attribute features construction

In biomolecular networks, the relationships among nodes of the same category exhibit a high degree of intricacy [23]. When considering two nodes of the same category, their interactions with nodes of different categories exhibit partial overlap. The extent of shared elements is positively correlated with the functional similarity between two nodes, which can also be assessed by measuring their distance within the same category. To effectively capture these functional attributes, GraCMI employs Gaussian interaction profile kernel (GIPK) and Jaccard similarity to compute the functional similarity [24]. The similarity features are further processed through a Sparse Autoencoder (SAE) to eliminate noise and enhance the expressiveness of the learned feature representations.

Calculate functional similarity. We denote CM as the circRNA-miRNA relationship matrix, where rows represent circRNAs and columns represent miRNAs. If there is an association between circRNA c_i and miRNA m_j, then $CM(c_i, m_j) = 1$, otherwise, it is 0. The GIPK similarity between circRNA c_i and circRNA c_j, denoted as $K(c_i, c_j)$, is calculated as follows:

$$K(c_i, c_j) = \exp(-\gamma_c \|CM(c_i) - CM(c_j)\|^2) \tag{1}$$

where $CM(c_i)$ and $CM(c_j)$ represent the row in the CM corresponding to circRNA c_i and circRNA c_j, respectively. γ_c is the bandwidth parameter of the GIPK function, and its calculation formula is as follows:

$$\gamma_c = \frac{1}{\frac{1}{n_c}\sum_{i=1}^{n_c} \|CM(c_i)\|^2} \tag{2}$$

where n_c represents the number of circRNAs. The process of computing the GIPK similarity for miRNA is analogous to that of circRNA.

To address the issue of node sparsity in molecular heterogeneous networks, we introduce the Jaccard similarity, an index used to measure the similarity between two sets [25]. It focuses solely on whether nodes in the sets exist or are missing and is thus not influenced by the number and values of nodes in the sets. The Jaccard similarity between circRNA c_i and circRNA c_j, denoted as $J(c_i, c_j)$, is calculated as follows:

$$J(c_i, c_j) = \frac{|M_i \cap M_j|}{|M_i \cup M_j|} \tag{3}$$

where M_i and M_j represent the collections of miRNAs related to circRNA c_i and circRNA c_j, respectively. The process of computing the Jaccard similarity for miRNA is analogous to that of circRNA.

We define the circRNA similarity matrix as $CS = (K(c_i, c_j) + J(c_i, c_j))/2$ and the miRNA similarity matrix as $MS = (K(m_i, m_j) + J(m_i, m_j))/2$. The final similarity matrix S is defined as follows:

$$S = \begin{bmatrix} CS & CM^T \\ CM & MS \end{bmatrix} \tag{4}$$

Reconstruct similarity features. GraCMI employs a Sparse Autoencoder (SAE) to reconstruct and enhance the representation of similarity features. The SAE, as an extension of the conventional autoencoder, incorporates a sparsity constraint into its encoding process, wherein only a small subset of neurons are activated while the majority remain inactive. This constraint facilitates the extraction of essential features from the input data more effectively. The SAE consists of three layers: an input layer, a hidden layer, and an output layer. The input layer comprises neurons equal in number to the dimensionality of the input feature space, with each neuron representing a specific feature. Its primary role is to receive original feature representations and forward them to the hidden layer for further encoding.

The hidden layer is the core component of the SAE and typically contains multiple neurons. By introducing a sparsity penalty term in the hidden layer, the neurons can identify key features within the data and maintain sparsity in their representations. The output F_h of the hidden layer is calculated as follows:

$$F_h = f(W_e X(i) + b_e) \tag{5}$$

where $X(i)$ represents the i-th element of the input vector, W_e and b_e represent the weight matrix and bias vector. f represents the activation function. The formula for calculating the sparse penalty term P_s is as follows:

$$P_s = \sum_{u=1}^{v} KL(\rho \| \hat{\rho}_u) \tag{6}$$

where v represents the number of neurons in the hidden layer, and ρ represents the desired sparsity. KL is the Kullback-Leibler divergence, and the smaller its value, the better the sparsity of the hidden layer [26]. $\hat{\rho}_u$ denotes the average activation of the u-th neuron in the hidden layer, and it is calculated as follows:

$$\hat{\rho}_u = \frac{1}{n}\sum_{i=1}^{n} \left[a_u(X(i))\right] \tag{7}$$

where n is the number of training samples, a_u represents the activation of the u-th neuron when the input sample is $X(i)$.

To enhance the sparsity of the hidden layer, we incorporate the KL divergence into the loss function and minimize it using gradient descent algorithm. The definition is as follows:

$$KL(\rho \| \hat{\rho}_u) = \rho \log \frac{\rho}{\hat{\rho}_u} + (1-\rho)\log \frac{1-\rho}{1-\hat{\rho}_u} \tag{8}$$

To maximize the capture of intrinsic features in molecules, we employ the Rectified Linear Unit (ReLU) as the activation function, along with an L1 regularization term to learn the sparsity of the representation.

In the output layer, we leverage the sigmoid function as the activation function to prevent issues such as exploding or vanishing values. The similarity matrix S is reconstructed via the SAE to refine the molecular attribute feature representation.

3.3. Multi-hop graph structural modeling

In traditional methods for extracting structural features, it is challenging to capture the global characteristics of a network [27]. Although existing graph embedding methods have achieved notable progress, they remain limited in effectively modeling complex and heterogeneous network data. For instance, Large-scale Information Network Embedding (LINE) utilizes a loss function based on first- and second-order proximity, which constrains its ability to capture long-range dependencies in graph representations [28]. In contrast, Grarep captures multi-hop structural features by computing transition matrices and utilizing a customized loss function.

Firstly, the transition probability matrix X^k is obtained for each value of k, $k = 1, 2, \dots, n$, which is calculated as follows:

$$X^k_{i,j} = \max\left(Y^k_{i,j}, 0\right) \tag{9}$$

Subsequently, singular value decomposition is applied to these probability matrices to derive k-step representations for every node. The matrix X^k is decomposed as follows:

$$X^k \approx X^k_d = U^k_d \Sigma^k_d \left(V^k_d\right)^T = W^k C^k \tag{10}$$

where Σ^k_d is the matrix composed of the first d singular values. U^k_d and V^k_d are the first d columns of U^k and V^k, respectively. Here W^k and C^k are calculated as follows:

$$W^k = U^k_d \left(\Sigma^k_d\right)^{\frac{1}{2}} \tag{11}$$

$$C^k = \left(\Sigma^k_d\right)^{\frac{1}{2}} V^{k^T}_d \tag{12}$$

Finally, concatenate all k-step representations to form a global representation.

The calculation of the k-step loss function for Grarep is outlined as follows:

$$L_k = \sum_{w \in V} L_k(w) \tag{13}$$

where $L_k(w)$ represents the k-step loss for node w, and its computation is defined as follows:

$$L_k(w) = \left(\sum_{c \in V} p_k(c \mid w) \log \sigma(\vec{w} \cdot \vec{c})\right) + \lambda \mathbb{E}_{c' \sim p_k(V)}\left[\log \sigma\left(-\vec{w} \cdot \vec{c'}\right)\right] \tag{14}$$

where $p_k(c \mid w)$ represents the k-step transition probability from the current vertex w to the context vertex c, and $p_k(V)$ represents the corresponding distribution over vertices in the graph. The c' indicates a vertex sampled from the negative sample set, where λ is a hyperparameter that controls the number of negative samples. Similarly, w' represents a negatively sampled starting vertex, and $q(w')$ denotes the probability of selecting w' as the initial vertex in the path. We assume $1 \le k \le n$, and as k increases, the transition probabilities gradually converge to a stable distribution. The distribution $p_k(c)$ is computed as follows:

$$p_k(c) = \sum_{w'} q(w') p_k(c \mid w') = \frac{1}{N}\sum_{w'} A^k_{w',c} \tag{15}$$

$\mathbb{E}$ denotes the expectation when c is drawn from the distribution $p_k(V)$, and is computed as follows:

$$\mathbb{E}_{c'\sim p_k(V)}\left[\log\sigma\left(-\vec{w}\cdot\vec{c'}\right)\right] = p_k(c)\cdot\log\sigma(-\vec{w}\cdot\vec{c}) + \sum_{c'\in V\setminus\{c\}} p_k\left(c'\right)\cdot\log\sigma\left(-\vec{w}\cdot\vec{c'}\right) \tag{16}$$

By defining the loss function, Grarep can effectively aggregate the global structural features of molecules within the network. Detailed algorithmic descriptions are provided in Algorithm 1.

Algorithm 1 Grarep.

Input
Adjacency matrix S on graph
Maximum transition step K
Log shifted factor β
Dimension of representation vector d
1. Get k-step transition probability matrix A^k
  Compute $A = D^{-1}S$
  Calculate $A^1, A^2, \dots, A^K$ respectively
2. Get each k-step representations
For $k = 1$ **to** K
  2.1 Get positive log probability matrix
  calculate $\Gamma_1^k, \Gamma_2^k, \dots, \Gamma_N^k$ ($\Gamma_j^k = \sum_p A_{p,j}^k$) respectively
  calculate $\{X_{i,j}^k\}$
  $X_{i,j}^k = \log\left(\frac{A_{i,j}^k}{\Gamma_j^k}\right) - \log(\beta)$
  assign negative entries of X^k to 0
  2.2 Construct the representation vector W^k
  $[U^k, \Sigma^k, (V^k)^T] = \mathrm{SVD}(X^k)$
  $W^k = U_d^k(\Sigma_d^k)^{\frac{1}{2}}$
End For
3. Concatenate all the k-step representations
$W = [W^1, W^2, \dots, W^K]$
Output
Matrix of the graph representation W

3.4. Prediction and cross-validation

The eXtreme Gradient Boosting (XGBoost) is a variant of the Gradient Boosting Decision Tree (GBDT) [29]. To make the classifier aware of the sparsity of input data, XGBoost adds a default direction at each tree node. We treat CMIs prediction as a binary classification problem, employing a 5-fold cross-validation (CV) to split the dataset into training and testing sets. The molecular fusion features are paired with corresponding labels and fed into the XGBoost classifier to predict potential associations between circRNAs and miRNAs.

3.5. Implementation details and computational analysis

Experimental setup. For computational resources, all experiments were conducted on a machine equipped with an NVIDIA GeForce RTX 4070 GPU, an Intel Core i9-14900HX processor, 32 GB RAM, and a 64-bit operating system based on x64 architecture. The code was implemented in Python 3.9. In GraCMI, we set k-step to 4 and the embedding dimension to 128 to capture multi-hop structure and balance representation. α is set to 0.5 to ensure stable scaling during singular value decomposition. The rdpie, also set to 0.5, is used to smooth the similarity matrices. The clf-ratio is set to 0.9 to maintain a 9:1 ratio between positive and negative samples.

Model complexity. The computational complexity of GraCMI, which primarily consists of the Grarep and SAE modules, is determined by the respective computational demands of these two components. Specifically, Grarep exhibits a time complexity of $O(K\cdot n\cdot m + n^2\cdot d)$ and a space complexity of $O(m + n^2 + n\cdot d)$, where K denotes the number of propagation steps, n and m represent the number of nodes and edges respectively, and d is the embedding dimension. In comparison, the SAE module has a time complexity of $O(n^3)$ and a space complexity of $O(n^2)$.

4. Results

4.1. Performance evaluation and prediction results

In a 5-fold CV experiment, to assess the performance of GraCMI, we adopted several evaluation criteria on the CMI-755 dataset, including accuracy (Acc.), precision (Prec.), recall (Rec.), specificity (Spec.), F1 score, Matthews correlation coefficient (MCC), area under receiver operating characteristic (ROC) curve (AUC), and area under precision-recall (PR) curve (AUPR). The detailed prediction results are shown in Table 2, the average Acc., Prec., Spec., Rec., F1, MCC, AUC, and AUPR achieved by GraCMI in 5-fold CV experiments are 73.97 %, 78 %, 66.89 %, 81.06 %, 71.86 %, 48.57 %, 0.8319, and 0.8285, respectively. Furthermore, we plotted the ROC and PR curve, as illustrated in Fig. 3.

4.2. Generalization ability

To assess the generalization capability of GraCMI, we conducted testing on multiple datasets. Specifically, we performed 5-fold CV experiments on the CMI-755, CMI-9589 and CMI-9905 datasets as illustrated in Fig. 4. The results demonstrate that GraCMI achieves superior performance on the CMI-9589 and CMI-9905. Due to the sparse data and the lower richness of extracted features in CMI-755, GraCMI exhibited reduced predictive performance on the dataset. It is worth noting that the CMI-755 dataset is cancer-related and experimentally validated, making the predictions on it practically meaningful. Overall, the strong generalization ability of GraCMI is well demonstrated across the prediction results of these three datasets.

4.3. Ablation experiments

GraCMI predicts potential CMIs by learning comprehensive feature representations of molecules in the network from multiple perspectives. To unravel the distinct contributions of these features to the proposed model, we conducted a series of ablation experiments. Specifically, we sequentially removed the molecular attribute features (W/O Attribute) and multi-hop structural features (W/O Multi-hop) to assess their impact on the experimental results.

Table 2
The 5-fold CV results of GraCMI on the CMI-755 dataset.

Test set	Acc. (%)	Prec. (%)	Rec. (%)	Spec. (%)	F1 (%)	MCC (%)	AUC	AUPR
1	79.80	80.82	78.15	81.46	79.46	59.64	0.8808	0.8775
2	73.84	78.57	65.56	82.12	71.48	48.35	0.8451	0.8242
3	78.15	85.12	68.21	88.08	75.74	57.44	0.8750	0.8625
4	68.54	74.56	56.29	80.79	64.15	38.25	0.7854	0.7886
5	69.54	70.92	66.23	72.85	68.49	39.16	0.7733	0.7897
Average	73.97	78.00	66.89	81.06	71.86	48.57	0.8319	0.8285
Std	4.48	4.92	6.97	4.86	5.36	8.90	0.0448	0.0365

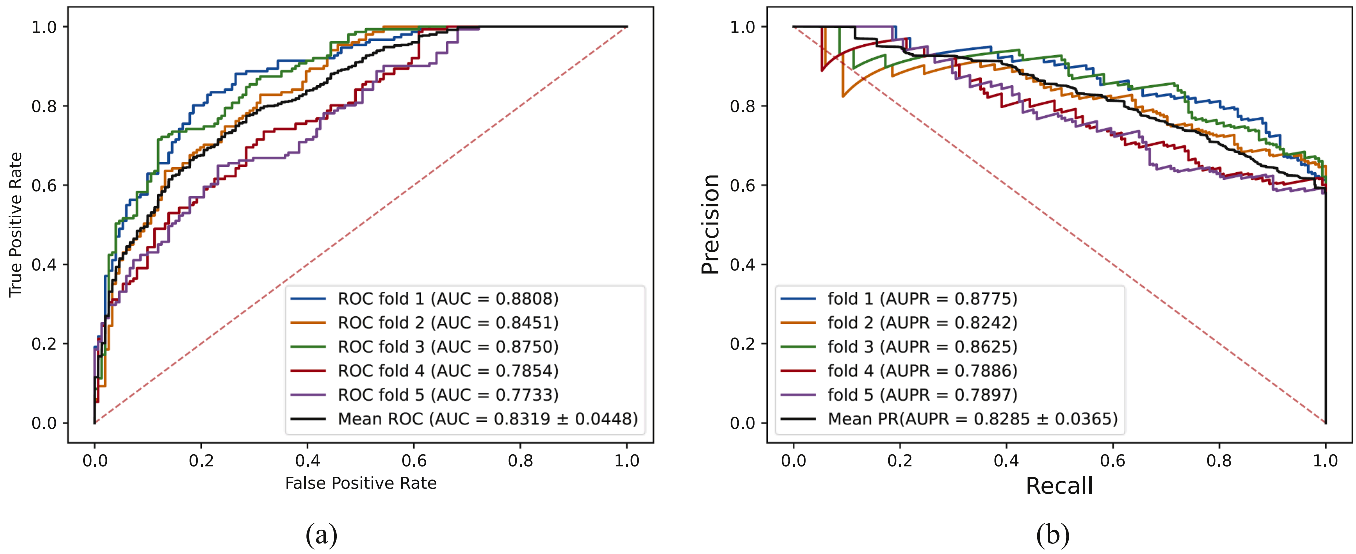

Fig. 3. The ROC curve (a) and PR curve (b) of GraCMI on the CMI-755 dataset.

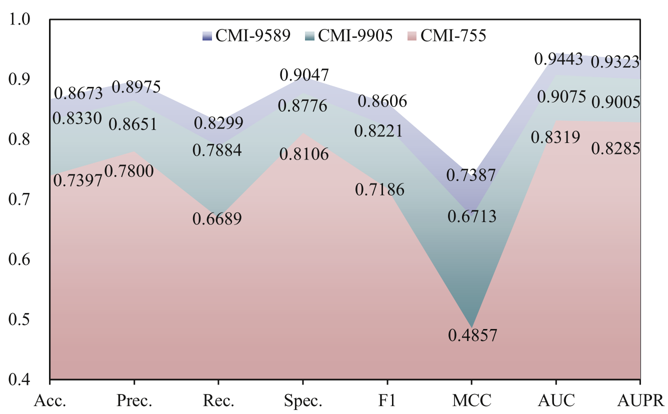

Fig. 4. The 5-fold CV results performed by GraCMI on the different datasets.

The prediction results under the 5-fold CV are shown in Fig. 5. Deleting the multi-hop structural features of the molecule has a greater impact on the prediction results, while deleting the molecular attribute features has a smaller impact. Notably, the combination of both multi-hop structural features and molecular attribute features achieves the best overall prediction performance. The ablation experiment not only proves the effectiveness of feature fusion but also enhances model interpretability.

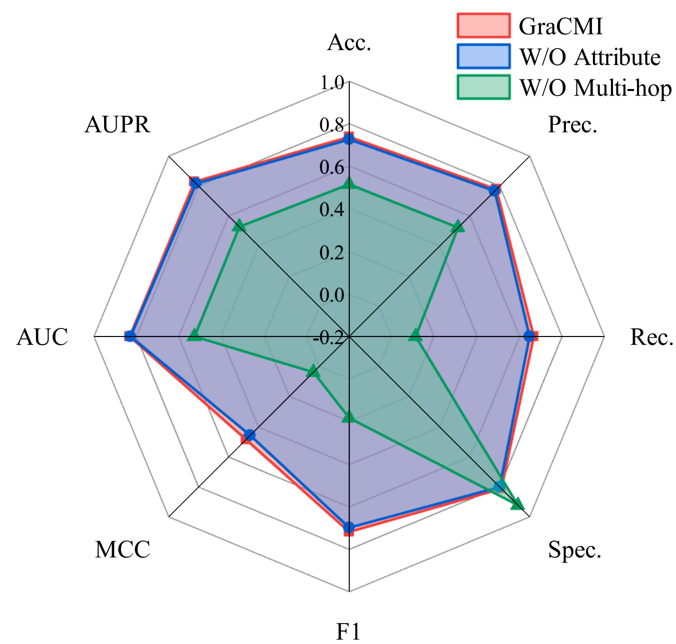

Fig. 5. The 5-fold CV results of GraCMI with W/O Attribute and W/O Multi-hop features.

4.4. Impact of different k-step

GraCMI learns representations of the circRNA-miRNA association graph based on the multi-hop principle, traversing subgraphs by defining the neighborhood range of the target nodes to extract structural information. The range of neighboring nodes has a significant impact on the target node representation [30]. Therefore, analyzing and determining the optimal neighborhood range can optimize prediction performance. We utilize GraCMI to extract structural features of target nodes based on different k-step neighborhood information. Specifically, we first aggregate learning on the 1, 2, 3, 4, and 5-step neighborhood information of the target nodes respectively. Then, the obtained graph structure features are fused with node attribute features. Finally, the fused features are input into an XGBoost classifier to predict unknown CMIs.

The experimental results of different k-step are shown in Table 3. When the neighborhood range k of the target node is 4, the model achieves the best performance. The performance is poorest when k is 1, suggesting that a too small neighborhood range makes it difficult for the model to aggregate key information relevant to the target node, resulting in insufficient node expression. Conversely, when k is 5, irrelevant node information may be aggregated due to the excessively large neighborhood range, negatively impacting the representation of the target node.

4.5. Comparison of graph representation methods

The ablation experiment results demonstrate that the molecular multi-hop graph structure features contribute substantially to the prediction of CMIs. Therefore, we compare GraCMI with several traditional graph embedding methods, including Structural Deep Network Embedding (SDNE) [31], DeepWalk [32], and Large-scale Information Network Embedding (LINE) [28]. In addition, we also compare two recent noise-resilient graph embedding methods, MetaGC [33] and ADA-GAD [34].

Table 3
The results of GraCMI on different k-step.

k-step	Acc. (%)	Prec. (%)	Rec. (%)	Spec. (%)	F1 (%)	MCC (%)	AUC	AUPR
1	68.34	73.15	58.94	77.75	65.10	37.60	0.7501	0.7547
2	70.60	75.16	61.59	79.60	67.64	41.93	0.8183	0.8056
3	70.86	74.47	63.71	78.01	68.51	42.30	0.8175	0.8236
4	**73.97**	**78.00**	**66.89**	**81.06**	**71.86**	**48.57**	**0.8319**	**0.8285**
5	72.19	76.02	65.03	79.34	69.96	44.95	0.8182	0.8239

- SDNE leverages an autoencoder framework to jointly optimize first-order and second-order similarities, thereby preserving local structural information in the learned representations.
- DeepWalk transforms the graph structure into node sequences via random walks and learns node representations by capturing co-occurrence relationships among nodes.
- LINE generates node embeddings by preserving both first-order and second-order proximity information, effectively capturing local network structures.
- MetaGC adopts a meta-learning approach to assign adaptive weights to node pairs, enhancing informative interactions while mitigating the influence of noisy links.
- ADA-GAD constructs refined graph views through unsupervised augmentation and retrains the decoding module, promoting stability by incorporating node-level anomaly distribution constraints.

The experimental results of comparing different graph embedding methods using 5-fold CV are presented in Table 4. GraCMI shows superior performance across most evaluation metrics. Among the classical graph embedding methods, SDNE performs relatively well, while DeepWalk shows slightly inferior results, and LINE performs the worst, indicating its limited capacity to model the long-range dependencies of the CMI network. In contrast, the denoising-based methods MetaGC and ADA-GAD demonstrate strengths in recall and F1, respectively, but still fall short of GraCMI in terms of overall balance. This can be attributed to the incorporation of denoising mechanisms in these two methods, which enhance their ability to mitigate the impact of noisy edges and improve robustness. However, despite these advantages, the integration of multi-hop structural information and advanced denoising techniques in GraCMI enables the model to achieve a more comprehensive and balanced representation.

In the binary classification tasks, the quality of local feature clustering contributes significantly to revealing underlying patterns and relationships among nodes. When similar features are grouped, classifiers can more easily recognize and utilize these clusters, enhancing predictive performance. t-SNE (t-distributed Stochastic Neighbor Embedding) effectively preserves structural relationships between nodes during visualization, providing a clearer representation of the clustering patterns in high-dimensional feature spaces [35]. To further evaluate the clustering capability of graph embedding methods, we employed t-SNE to project the high-dimensional features extracted by GraCMI, SDNE, DeepWalk, and LINE into a lower-dimensional space. Specifically, the features with node category labels are input into t-SNE and reduced to 2 dimensions, resulting in the visualization of these two-dimensional feature representations, as depicted in Fig. 6.

Consistent with the findings in Table 4, most of the nodes extracted by GraCMI which belong to the same class are clustered together, demonstrating its strong feature representation capability. The clustering performance of LINE is relatively poor, with all nodes exhibiting a random distribution. While SDNE and DeepWalk closely cluster nodes of the same class, they do not exhibit distinct boundary lines. In addition, we also visualized the score distributions predicted by GraCMI, SDNE, DeepWalk, and LINE, as shown in Fig. 7. Overall, the score distribution of GraCMI demonstrates a stronger polarization trend, effectively separating associated from non-associated pairs. This further validates its superior ability to capture complex relationships among biomolecules.

4.6. Impact of different embedding dimensions

GraCMI leverages matrix factorization techniques to extract higher-order structural features through a multi-hop mechanism. To investigate the effect of structural feature dimensionality on model performance, we conducted comparative experiments with embedding sizes of 32, 64, 128, and 256. The specific results are shown in Fig. 8, where the solid line represents the average value of the 5-fold CV and the dotted line represents the standard deviation.

Experimental results demonstrate that GraCMI exhibits certain performance variations under different embedding dimensionalities. Specifically, when the embedding dimension is set to 128, the model achieves relatively superior results across most evaluation metrics. The model achieves the second-best performance at 32 dimensions with balanced results across metrics. When the dimension increases to 256, several metrics show a decline, suggesting that excessive dimensionality may introduce redundant information. It is worth noting that the standard deviations at 64 and 128 dimensions are relatively higher, indicating reduced stability of the model under these settings. Taking both predictive performance and stability into consideration, the 128-dimensional embedding can be regarded as a more suitable configuration for GraCMI.

4.7. Optimal classification strategy

To take full advantage of the extracted fused features, GraCMI employs the XGBoost classifier to score each potential circRNA-miRNA pair. In this section, we compare XGBoost with several mainstream classifiers, including Light Gradient Boosting Machine (LightGBM), Adaptive Boosting (AdaBoost), SVM (Support Vector Machines), RF (Random Forest), Multilayer Perceptron (MLP), and Convolutional Neural Network (CNN) [36,37]. The comparative results using 5-fold CV are illustrated in Table 5 and Fig. 9.

Compared to other classifiers, XGBoost achieves superior performance across multiple metrics, demonstrating a strong ability to

Table 4
The results of GraCMI on different graph embedding methods.

Methods	Acc. (%)	Prec. (%)	Rec. (%)	Spec. (%)	F1 (%)	MCC (%)	AUC	AUPR
GraCMI	**73.97**	**78.00**	66.89	**81.06**	71.86	**48.57**	**0.8319**	**0.8285**
SDNE	61.59	63.22	55.76	67.42	59.20	23.39	0.6683	0.7010
DeepWalk	59.80	61.08	53.64	65.96	57.07	19.75	0.6387	0.6211
LINE	52.25	52.63	46.62	57.88	49.31	4.57	0.5104	0.5285
MetaGC	71.72	69.01	78.68	64.77	73.35	44.22	0.7824	0.7340
ADA-GAD	70.07	66.04	**82.78**	57.35	**73.43**	41.59	0.7723	0.7427

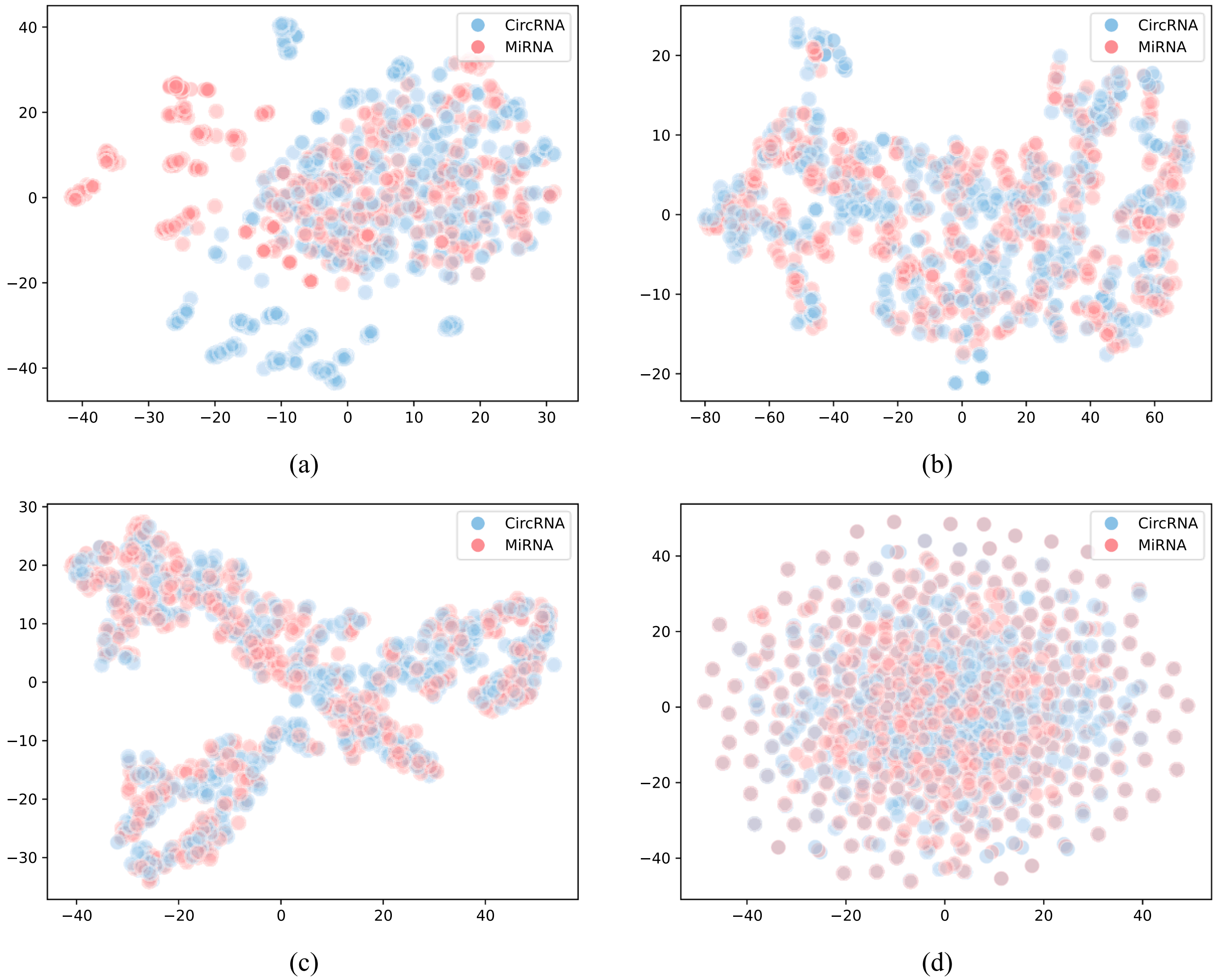

Fig. 6. Visual analysis of different graph embedding methods, GraCMI (a), SDNE (b), DeepWalk (c), and LINE (d).

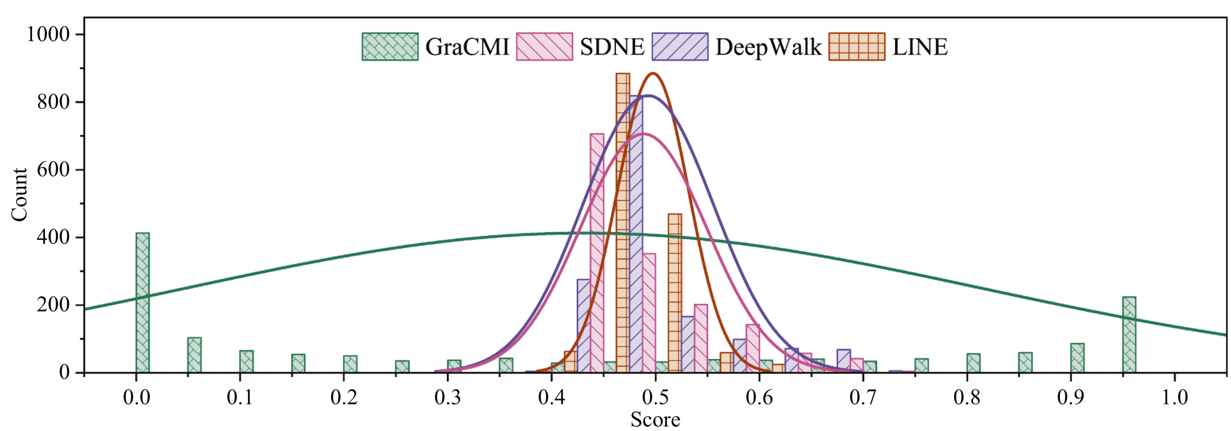

Fig. 7. Distribution of prediction scores for GraCMI and baseline models.

capture underlying molecular relationships. Additionally, the other classifiers also exhibit competitive results, suggesting that the denoising of molecular attribute features, along with multi-hop structural modeling, effectively contributes to the enhancement of molecular representations.

4.8. Comparison with other latest models

To evaluate the performance of GraCMI, we compared it with the latest models on three datasets. AUC, AUPR, Acc., and Prec. are used as indicators to evaluate the model. Specifically, on CMI-755, we compared

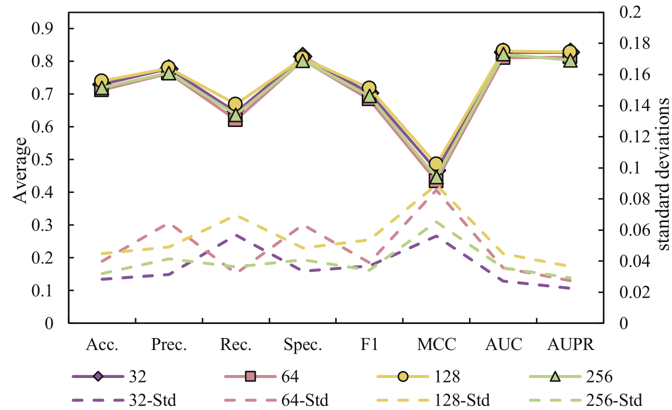

Fig. 8. Performance comparison of structural features with different embedding dimensions.

Table 5
The results of GraCMI on different classifiers.

Classifier	Acc. (%)	Prec. (%)	Rec. (%)	Spec. (%)	F1 (%)	MCC (%)	AUC	AUPR
XGBoost	**73.97**	**78.00**	**66.89**	**81.06**	**71.86**	**48.57**	**0.8319**	**0.8285**
LightGBM	65.03	65.30	62.25	67.81	63.56	30.19	0.7189	0.7033
AdaBoost	50.07	50.32	45.83	54.30	47.93	0.18	0.4697	0.5325
SVM	68.74	73.91	58.01	79.47	64.71	38.53	0.7476	0.7831
RF	66.03	71.72	53.91	78.15	60.70	33.50	0.7577	0.7881
MLP	68.94	71.90	62.12	75.76	66.45	38.37	0.7974	0.8229
CNN	66.36	69.99	55.36	77.35	60.80	33.66	0.7151	0.6919

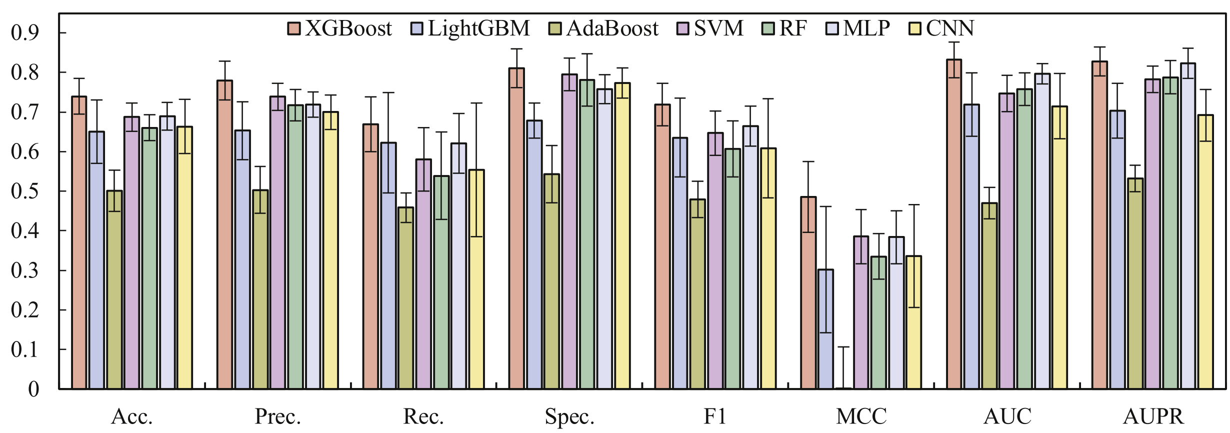

Fig. 9. Performance comparison of GraCMI with different classifiers.

the proposed model with KS-CMI [38] and BCMCMI [20], on CMI-9589 with JSNDCMI[14], KS-CMI, and BCMCMI, and on CMI-9905 with JS-NDCMI and BCMCMI.

- KS-CMI learns feature representation by constructing the circRNA-miRNA-cancer network and integrating balance theory to enrich the behavioral relationship features between molecules.
- JSNDCMI addresses the challenge of sparse network representation by integrating functional similarity and local topological similarity, while utilizing a Denoising Autoencoder (DAE) to enhance feature learning.
- BCMCMI extracts semantic features of molecules using the BERT and constructs a heterogeneous network based on cosine similarity, employing Metapath2vec to capture topological features across different patterns.

The detailed comparison results are shown in Table 6, some of which are derived from publicly available literature, and others from our reproduction of published model codes. GraCMI demonstrates remarkable performance across multiple evaluation metrics on three datasets, underscoring its significant advantage over existing models.

Table 6
Comparison with state-of-the-art methods on the three datasets.

Dataset	Model	AUC	AUPR	Acc. (%)	Prec. (%)
CMI-755	KS-CMI	0.8132	0.7936	73.57	73.63
	BCMCMI	0.7240	0.7286	64.97	66.42
	GraCMI	**0.8319**	**0.8285**	**73.97**	**78.00**
CMI-9589	JSNDCMI	0.9061	0.9051	83.18	**90.43**
	KS-CMI	0.9180	0.9181	83.50	83.55
	BCMCMI	0.9372	0.9297	86.25	84.14
	GraCMI	**0.9443**	**0.9323**	**86.73**	89.75
CMI-9905	JSNDCMI	0.9007	0.8999	82.32	82.32
	BCMCMI	0.9041	0.8990	83.16	80.83
	GraCMI	**0.9075**	**0.9005**	**83.30**	**86.51**

4.9. Case studies

In-depth investigation into the association between CMIs and diseases can significantly promote the elucidation of the underlying mechanisms of disease pathogenesis [39]. To further assess the prac-

Table 7
The top 10 circRNAs and miRNAs with predicted scores.

Rank	CircRNA	MiRNA	Cancer	Evidence (PMID)
1	has_circ_0001955	miR-145-5p	hepatocellular carcinoma	31822654
2	hsa_circ_0043949	miR-4443	N/A	N/A
3	hsa_circ_0001946	miR-671-5p	glioblastoma	30663767
4	hsa_circ_0000376	miR-200	N/A	N/A
5	circCEP128	miR-411-5p	N/A	N/A
6	hsa_circ_0035483	hsa-miR-335	renal clear cell carcinoma	31492499
7	circ-ITCH	miR-22	papillary thyroid cancer	30190130
8	hsa_circ_0007294	miR-148a-3p	triple-negative breast cancer	30454010
9	hsa_circ_0002211	miR-31-5p	colorectal cancer	32141542
10	hsa_circ_0001846	miR-661	triple-negative breast cancer	30314706

ticality of GraCMI, we conducted case studies. Specifically, we utilized the cancer-related CMI-755 dataset as a training sample and employed GraCMI to predict potential relationships between circRNAs and miRNAs. In Table 7, we present the top 10 circRNA-miRNA pairs according to prediction scores, among which 7 pairs were confirmed and supported by relevant literature. Additionally, we compiled information on the cancers associated with these confirmed interacting pairs. In conclusion, the case studies indicate that GraCMI can furnish researchers with highly relevant candidate interaction pairs, thereby narrowing the scope of wet lab experiments and expediting understanding of disease mechanisms.

5. Conclusion

In this paper, we propose a novel model named GraCMI, which predicts potential CMIs based on multi-hop graph structural modeling. GraCMI captures global structural information by performing matrix decomposition on multi-order adjacency matrices. To address the noise issue in biological attribute information, GraCMI employs the SAE to denoise the integrated attribute features constructed based on functional similarity. Multiple comparative experiments and case studies demonstrated the reliability of GraCMI in predicting potential CMIs. Furthermore, we have developed a computational platform available at http://39.106.153.201/GraCMI/home.html to facilitate collaboration and communication in the field and to provide researchers with easier access to relevant resources.

Despite the promising results achieved by GraCMI, several limitations remain. First, the model relies on randomly generated negative samples, which may lack biological relevance. We plan to incorporate molecular similarity principles to construct more biologically meaningful negative samples. Second, the $O(n^3)$ time complexity of the denoising module may hinder scalability to large biological networks. To address this limitation, we will explore lightweight denoising strategies that reduce computational overhead while preserving predictive performance. Moreover, although GraCMI has been validated on cancer-related CMIs, extending it to other RNA types, such as long non-coding RNAs (lncRNAs) or messenger RNAs (mRNAs), may require algorithmic adaptation to accommodate distinct molecular characteristics. Future work will focus on developing specialized feature extraction methods to enhance extensibility across diverse biomedical domains.

CRediT authorship contribution statement

Mengmeng Wei: Data curation, Writing – original draft; **Lei Wang:** Writing – review & editing, Funding acquisition, Investigation; **Xiaorui Su:** Methodology, Conceptualization; **Bowei Zhao:** Validation, Resources; **Zhuhong You:** Writing – review & editing, Supervision.

Data availability

Data will be made available on request.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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